



Vitamin D insufficiency and deficiency in children and adolescents

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INTRODUCTION

Vitamin D is an essential nutrient that plays an important role in calcium homeostasis and bone health.

Severe deficiency of vitamin D causes rickets in infants and children, and osteomalacia in all age groups. Severe vitamin D deficiency may also be associated with hypocalcemia, which may cause tetany or seizures. These disorders occur with the highest frequency among children in malnourished populations living in resource-limited countries. Rickets also occurs in children in resource-rich countries if sufficient vitamin D intake is not ensured through the use of supplements and fortified foods, particularly if exposure to sunlight is limited, and in children with chronic illnesses. The clinical evaluation and treatment of a child with rickets is discussed separately. (See "[Overview of rickets in children](#)" and "[Etiology and treatment of calcipenic rickets in children](#)", [section on 'Nutritional rickets'](#).)

The clinical consequences of mild vitamin D deficiency are less well established. However, chronically low vitamin D levels are associated with the development of low bone mineral density and other measures of reduced bone health, even in the absence of rickets.

The diagnosis, causes, and prevention of vitamin D deficiency in children, and the treatment of vitamin D deficiency in the absence of rickets will be reviewed here. The causes and treatment of vitamin D deficiency in adults are discussed in separate topic reviews. (See "[Causes of vitamin D deficiency and resistance](#)" and "[Vitamin D deficiency in adults: Definition, clinical manifestations, and treatment](#)".)

METABOLISM AND FORMS OF VITAMIN D

Vitamin D is a prohormone that is synthesized in the skin after exposure to ultraviolet radiation, or absorbed from food sources or supplements. The prohormone is then serially converted to the metabolically active form in the liver and subsequently the kidneys ([figure 1](#)). (See "[Overview of vitamin D](#)", [section on 'Metabolism'](#)".)

The main forms of vitamin D are:

- **Cholecalciferol**, or vitamin D₃, is the form of vitamin D found in animal products and some vitamin D supplements. It is formed when ultraviolet B (UVB) radiation (wavelength 290 to 315 nm) converts 7-dehydrocholesterol in epidermal keratinocytes and dermal fibroblasts to pre-vitamin D, which subsequently isomerizes to [vitamin D3](#).
- **Ergocalciferol**, or vitamin D₂, is the form of vitamin D found in plant dietary sources and in most vitamin D supplements. It is formed when ergosterol in plants is exposed to irradiation.
- **Calcidiol** (25-hydroxyvitamin D [25OHD]), is the storage form of vitamin D. It is formed in the liver after vitamin D (cholecalciferol produced in the skin or ingested, or ergocalciferol ingested) is bound to vitamin D-binding protein (VDBP) and transported to the liver, where it undergoes 25-hydroxylation to form 25OHD.
- **Calcitriol** (1,25-dihydroxyvitamin D or 1,25[OH]₂D), is the active form of vitamin D. It is formed in the kidney, after 25OHD undergoes 1-alpha-hydroxylation to form 1,25-dihydroxyvitamin D. This process is driven by parathyroid hormone (PTH) and other mediators, including hypophosphatemia and growth hormone. Although kidney production of [calcitriol](#) regulates circulating levels of this active form of vitamin D, there are many sites of 1-alpha-hydroxylation, including lymph nodes, placenta, colon, breasts, osteoblasts, alveolar macrophages, activated

macrophages, and keratinocytes, suggesting an autocrine-paracrine role for 1,25-dihydroxyvitamin D at these sites [1].

SOURCES OF VITAMIN D

Very few foods naturally contain vitamin D; the main food source is oil-rich fish, such as salmon, mackerel, and herring, as well as liver and organ meats, and egg yolk. Few of these natural dietary sources are typically consumed by children consistently. The vitamin D content of breast milk is also low (see '[Exclusive breastfeeding](#)' below). Dermal synthesis is the major natural source of the vitamin. Individuals who do not have sufficient sun exposure, especially infants, require supplemental vitamin D from fortified foods or supplements. (See '[Prevention](#)' below.)

In the United States, milk, infant formula, breakfast cereals, and some other foods are fortified with vitamin D. In other parts of the world, cereals and bread products are often fortified with vitamin D. (See "[Overview of vitamin D](#)".)

TARGETS FOR VITAMIN D INTAKE

The following recommendations for vitamin D intake in healthy individuals are endorsed by the National Academy of Medicine (NAM) and the American Academy of Pediatrics (AAP) [2,3]:

- **Infants** (born at term) – 400 international units (10 micrograms) daily. Infants who are exclusively breastfed require vitamin D supplements to achieve this target, as do some formula-fed infants [1,4]. Guidance for supplementation is discussed below. (See '[Prevention in the perinatal period and in infants](#)' below.)

Supplementation for premature infants is discussed separately. (See "[Management of neonatal bone health](#)", [section on 'Vitamin D'](#)".)

- **Children 1 to 18 years of age** – 600 international units (15 micrograms) daily.

These doses are designed to maintain 25-hydroxyvitamin D (25OHD) levels >20 ng/mL (50 nmol/L) in most populations, which provides some margin of error to prevent rickets in infants and children [1,5,6].

For certain high-risk populations, even the recommended vitamin D dose of 600 international units (15 micrograms) daily for children may be insufficient to achieve the target 25OHD concentrations. These populations may require higher vitamin D doses to meet their predicted needs, based on population-specific data. For example, studies from Mediterranean and Middle Eastern regions indicate that as much as 2000 international units (50 micrograms) daily of vitamin D may be necessary to raise 25OHD levels to a healthy range (defined in this study as >30 ng/mL [75 nmol/L]) and maintain these levels over a year-long period [7,8]. The Canadian Pediatric Society recommends supplementation with 800 international units (20 micrograms) daily of vitamin D for breastfed infants living in northern communities during the winter [9]. The requirement for vitamin D may be even higher for infants of dark-skinned mothers (unless these mothers received adequate vitamin D supplementation through pregnancy) and those who live at high latitudes. (See '[Common postnatal risk factors](#)' below.)

There is limited evidence that fracture risk is associated with low levels of vitamin D intake. However, one large observational study found that vitamin D intake was associated with reduced risk of stress fractures among preadolescent and adolescent girls, particularly those participating in at least one hour/day of high-impact activity [10]. After adjusting for confounders, the risk of developing a stress fracture among girls in the highest quintile of vitamin D intake (mean intake 663 international units daily) was 50 percent lower than the risk in girls with the lowest quintile of vitamin D intake (mean intake 107 international units daily). Although this study does not establish a causal association between vitamin D intake and fracture risk, the findings lend support to the recommended daily intake level of 600 international units (15 micrograms) daily.

Within a given population, individuals with special risk factors may require doses of vitamin D that are higher than those recommended above. As examples, children with obesity and those on anticonvulsants, glucocorticoids, and medications for HIV infection may require much higher doses of vitamin D to maintain their vitamin D levels in the sufficient range (as much as 6000 international units [150 micrograms] daily). The dose should be guided by serum 25OHD levels, with a target concentration of at least 20 ng/mL (50 nmol/L). (See '[Obesity](#)' below and '[Medications](#)' below and '[Vitamin D replacement](#)' below.)

EPIDEMIOLOGY

Prevalence — In the United States, the overall prevalence of vitamin D deficiency or insufficiency (defined in these studies as 25-hydroxyvitamin D [25OHD] <20 ng/mL [50 nmol/L]) in the pediatric age range is approximately 15 percent, according to large population-based studies [11-13]. 25OHD levels <10 ng/mL (<25 nmol/L) were found in 1 to 2 percent of the pediatric population [11,12]. However, the prevalence varies considerably among different countries and subpopulations because of differences in risk factors, especially skin pigmentation, sun exposure, and dietary vitamin D intake. (See '[Pathogenesis and risk factors](#)' below.)

Associated conditions — Populations with higher rates of vitamin D deficiency also have higher rates of rickets and osteomalacia, which are the main pathophysiologic consequences. (See '[Clinical manifestations](#)' below and "[Etiology and treatment of calcipenic rickets in children](#)", [section on 'Nutritional rickets'](#).)

Epidemiologic studies suggest possible associations between vitamin D deficiency and a variety of conditions, but a causal relationship has not been established, and the mechanism for the associations are not clear. These include certain immunologic conditions such as multiple sclerosis [14], type 1 diabetes [15], rheumatoid arthritis [16], and inflammatory bowel disease [17], as well as mood disorders [18,19], cardiovascular disease [20], and cancers such as breast, prostate, and colon cancer [21-26]. In adolescents in the United States, low serum vitamin D levels are associated with increased risk of hypertension, hyperglycemia, and the metabolic syndrome, even after controlling for race/ethnicity, body mass index, socioeconomic status, and physical activity [27]. A higher risk of upper respiratory infections has also been associated with low vitamin D levels [28,29]. Associations with food allergies and asthma [30-36] and childhood dental caries [37-40] also have been reported in a variety of populations, although causal associations have not been established. (See "[Vitamin D and extraskeletal health](#)" and "[Risk factors for asthma](#)", [section on 'Maternal diet during pregnancy'](#).)

PATHOGENESIS AND RISK FACTORS

Vitamin D deficiency is common in infants who have dark skin pigmentation and those who are exclusively breastfed beyond three to six months of age, particularly if there are additional risk factors such as maternal vitamin D deficiency during pregnancy or prematurity. Vitamin D deficiency is also common among older children with dark skin

pigmentation and those who consume a diet with limited intake of vitamin D-fortified foods (eg, a diet low in dairy products, or some vegan and unusual diets), use anticonvulsant or antiretroviral medications, or have malabsorptive conditions, or obesity. Additional risk factors include residence at higher latitudes, winter season, and other causes of low sun exposure. Details about each of these risk factors are provided in the following sections.

Perinatal risk factors

Maternal vitamin D deficiency — Vitamin D is transferred from the mother to the fetus across the placenta, and reduced vitamin D stores in the mother are associated with lower vitamin D levels in the infant [41]. Vitamin D deficiency is particularly common in dark-skinned pregnant mothers, especially those living at higher latitudes and in the winter months [42-44]. In one report of pregnant women in the United States, most of whom had dark skin pigmentation, 50 percent had vitamin D insufficiency (25-hydroxyvitamin D [25OHD] <30 ng/mL) at delivery, despite maternal intake of approximately 600 international units/day (15 micrograms/day) through vitamin D supplements and milk [43]. Among their infants, 65 percent had 25OHD below this threshold at birth.

Prematurity — Vitamin D levels are particularly low in premature infants because they have less time to accumulate vitamin D from the mother through transplacental transfer [45]. The third trimester is a critical time for vitamin D transfer because this is when the fetal skeleton becomes calcified, requiring increased activation of 25OHD to 1,25-dihydroxyvitamin D in the maternal kidneys and placenta. Vitamin D deficiency in the mother during this period can cause fetal vitamin D deficiency and, in severe cases, fetal rickets. (See '[Prevention in the perinatal period and in infants](#)' below.)

Exclusive breastfeeding — The vitamin D content of breast milk is low (15 to 50 international units/L [0.4 to 1.2 micrograms/L]) even in a vitamin D-sufficient mother. Exclusively breastfed infants consuming an average of 750 mL of breast milk daily ingest only 10 to 40 international units/day (0.25 to 1 microgram/day) of vitamin D in the absence of sun exposure or supplement use [1,46,47]. The vitamin D content of breast milk is lower in mothers with dark skin or other causes of maternal vitamin D deficiency [48].

For most infants, exposure to sunlight is generally not a sufficient source of vitamin D. One study that included Black and White infants estimates that most breastfed infants

need to be exposed to sunlight for at least 30 minutes/week while wearing only a diaper in order to maintain 25OHD levels at >20 ng/mL (50 nmol/L) [41]. This amount of sun exposure is unlikely given more current recommendations to limit sun exposure in infants younger than six months old. (See '[Prevention in older children and adolescents](#)' below.)

Exclusive breastfeeding is an important risk factor for vitamin D deficiency and rickets. As an example, in a review of infants and children with rickets living in the United States, more than 95 percent were breastfed [49]. Similarly, in study from Alaska, 98 percent of infants with severe vitamin D deficiency (25OHD <10 ng/mL [25 nmol/L]) were exclusively breastfed [50].

Although vitamin D deficiency is uncommon in formula-fed infants because of the fortification of infant formulas, it can still occur if the infant had low vitamin D stores at birth because of maternal vitamin D deficiency and if the vitamin D content of the formula is insufficient to compensate for this. In one study, 50 percent of children with rickets at zero to five years old who presented with hypocalcemic convulsions had been formula fed [51]. (See '[Prevention in the perinatal period and in infants](#)' below.)

Common postnatal risk factors

Decreased nutritional intake — The primary natural (unfortified) dietary sources of vitamin D are oily fish (salmon, mackerel, sardines), cod liver oil, liver and organ meats, and egg yolk. These natural dietary sources are rarely consumed by children in sufficient amounts to maintain target 25OHD concentrations in the absence of other sources. The vitamin D content of breast milk is also low, as discussed above. (See '[Sources of vitamin D](#)' above.)

Because of the scarcity of natural dietary sources, vitamin D is fortified in many foods, particularly milk and milk products, orange juice, bread, and cereals. Infant formulas in the United States are required to contain 40 to 100 international units (1 to 2.5 micrograms) of vitamin D/100 kcal (usually providing at least 400 international units/L), and milk and orange juice that are labeled vitamin D-fortified are required to contain at least 400 international units (10 micrograms) of vitamin D per liter.

Despite these measures and the availability of vitamin D-fortified milk and other products, the dietary intake of vitamin D is often insufficient. Factors accounting for this include:

- Efforts to promote exclusive breastfeeding without coincident efforts to promote use of vitamin D supplements in these infants
- Inadequate consumption of fortified milk and dairy products by older children [52-54]
- Absence of standard regulations for vitamin D fortification across countries

Skin pigmentation and low sun exposure — Cutaneous vitamin D synthesis depends upon exposure to sunlight, specifically ultraviolet B (UVB), and this is reduced in children with increased skin pigmentation in whom melanin functions as a natural sunblock.

In individuals with light skin pigmentation, sufficient cutaneous vitamin D synthesis can be achieved by approximately 10 to 15 minutes of sun exposure (to the arms and legs; or hands, arms, and face) between 10:00 and 15:00 hours (10:00 AM and 3:00 PM), during the spring, summer, and fall [1]. Most Asian Indians require three times as much sun exposure as a light-skinned individual to achieve equivalent vitamin D concentrations, and individuals with very dark skin pigmentation (eg, some of those with African ancestry) require 6 to 10 times as much exposure as a light-skinned individual [55,56].

The overall prevalence of vitamin D deficiency among racial groups is generally correlated with differences in skin pigmentation. Accordingly, the highest prevalence of vitamin D deficiency rickets has been reported in Black children, followed by Asian Indians, and then White children [49,51,57]. Similarly, a national survey in the United States (the Third National Health and Nutrition Examination Survey [NHANES III]) indicates that Black non-Hispanic adolescents have 20 times the risk of having 25OHD levels of <20 ng/ mL (50 nmol/L) compared with White non-Hispanic adolescents [12]; the risk is higher in females compared with males.

The latitude of residence and season are also important determinants of cutaneous vitamin D synthesis [58-60]. During the winter months at high latitudes, there is greater scatter and absorption of UVB because of the oblique angle at which sunlight traverses the atmosphere and its longer path through the atmosphere. As a consequence, beyond a latitude of 40° and during winter, little or no UVB radiation reaches the surface of the earth. Therefore, while vitamin D deficiency is relatively uncommon at the end of the summer months, it is very common at the end of winter [58,59]. Even in the summer, use of sunblock can cause a persistence of low vitamin D levels [61].

In addition to the natural sunscreen of deeper skin pigmentation, UVB absorption is blocked by artificial sunscreens, and sunblock with a sun protection factor (SPF) of 30 can decrease vitamin D synthetic capacity by as much as 95 percent [62]. Other factors that can affect UVB exposure are altitude and cloud cover, and exposure is higher at greater altitudes and in areas where cloud cover is less. Staying indoors for long periods can also cause reduced vitamin D synthesis [63], and this can cause low vitamin D levels in disabled children and children who stay primarily indoors [64].

Obesity — An inverse association exists between obesity and 25OHD levels [27,65-67], that has been attributed to the sequestration of vitamin D in fat. Vitamin D requirements are thus higher in adolescents with obesity compared with normal weight adolescents. The clinical significance of low serum 25OHD levels in this group of patients is uncertain. (See "[Clinical evaluation of the obese child and adolescent](#)", [section on 'Laboratory studies'](#).)

Other risk factors — Less common but potentially strong risk factors for vitamin D deficiency include:

Medications

- Certain anticonvulsants and antiretroviral drugs used to treat HIV infection can precipitate vitamin D deficiency by enhancing catabolism of 25OHD and 1,25-dihydroxyvitamin D. (See "[Causes of vitamin D deficiency and resistance](#)", [section on 'Increased catabolism'](#).)
- Vitamin D requirements are higher in patients on glucocorticoids because they inhibit intestinal vitamin D-dependent calcium absorption. (See "[Overview of vitamin D](#)", [section on 'Deficiency and resistance'](#).)
- [Ketoconazole](#) and some other antifungal agents increase vitamin D requirements because they block 1-hydroxylation [68].

Malabsorption and other medical conditions — Conditions that impair fat absorption are associated with inadequate vitamin D absorption from the gut as this process is chylomicron-dependent. Rickets can therefore occur in children with celiac disease [69], inflammatory bowel disease, exocrine pancreatic insufficiency (as in cystic fibrosis), cholestasis, and following gut resection or bariatric surgery. (See "[Cystic fibrosis: Nutritional issues](#)", [section on 'Vitamin D'](#) and "[Causes of vitamin D deficiency and resistance](#)", [section on 'Gastric bypass'](#).)

Liver and kidney disease may be associated with deficient 25-hydroxylation and 1-hydroxylation, respectively, and therefore can cause rickets. (See "[Pediatric chronic kidney disease-mineral and bone disorder \(CKD-MBD\)](#)".)

Genetic disorders — Inherited diseases that cause vitamin D deficiency are [70]:

- 25-hydroxylase deficiency ([MIM #600081](#)), caused by mutations in *CYP2R1*, previously known as vitamin D-dependent rickets type 1B. This is a rare cause of vitamin D deficiency. Patients with heterozygous mutations have less severe clinical and biochemical features of vitamin D deficiency and a greater therapeutic response to high doses of vitamin D than those with homozygous mutations. The response to high vitamin D doses is only minimal in patients with homozygous mutations. (See "[Etiology and treatment of calcipenic rickets in children](#)", [section on '25-hydroxylase deficiency'](#).)
- 1-alpha-hydroxylase deficiency ([MIM #264700](#)), previously known as vitamin D-dependent rickets type 1A, caused by mutations in *CYP27B1*. The disorder has an autosomal pattern of inheritance and is characterized by early onset clinical and radiographic rickets with hypocalcemia, with normal levels of 25OHD and low levels of 1,25-dihydroxyvitamin D [71]. (See "[Etiology and treatment of calcipenic rickets in children](#)", [section on '1-alpha-hydroxylase deficiency'](#).)
- Hereditary resistance to vitamin D ([MIM #27740](#)), previously known as vitamin D-dependent rickets type 2, usually caused by mutations in the vitamin D receptor gene. Clinical features include alopecia and low calcium and phosphorus levels despite normal to high levels of both 25OHD and 1,25-dihydroxyvitamin D. (See "[Etiology and treatment of calcipenic rickets in children](#)", [section on 'Hereditary resistance to vitamin D'](#).)

Conditions with low 25-hydroxyvitamin D but without vitamin D deficiency — In certain conditions such as nephrotic syndrome, serum concentrations of 25OHD may be low because of low concentrations of vitamin D-binding protein (VDBP). However, this finding does not necessarily indicate vitamin D deficiency, because the serum **free** 25OHD concentration, which represents bioavailable 25OHD and is the physiologically important biomarker, may be normal. (See "[Endocrine dysfunction in the nephrotic syndrome](#)", [section on 'Vitamin D and calcium metabolism'](#).)

Genetic polymorphisms in VDBP vary markedly between racial and ethnic populations. Studies conflict regarding whether specific haplotypes, such as Gc1f, which is found more commonly in blacks, are associated with lower 25OHD concentrations compared with haplotypes GC1s and Gc2 because of lower VDBP levels. The racial differences in 25OHD concentrations reported by some studies may reflect the assay used. (See ["Vitamin D deficiency in adults: Definition, clinical manifestations, and treatment", section on 'Racial differences'.](#))

CLINICAL MANIFESTATIONS

Vitamin D deficiency causes rickets in growing children and osteomalacia in all age groups.

Biochemical changes — Vitamin D deficiency reduces intestinal calcium and phosphorus absorption.

Biochemical changes that characterize the different stages of vitamin D deficiency are outlined in the table ([table 1](#)). With increasing severity of vitamin D deficiency, 25-hydroxyvitamin D (25OHD) levels decrease, and parathyroid hormone (PTH) and alkaline phosphatase levels increase. 1,25-dihydroxyvitamin D levels initially increase in response to rising levels of PTH, but may subsequently decrease because of limited availability of its substrate, 25OHD. The increase in PTH mobilizes calcium from bone so that serum calcium levels remain normal or are only moderately decreased, despite reduced calcium absorption from the gut. Reduced gut absorption and rising PTH levels result in a progressive reduction in phosphorus levels.

Low phosphorus prevents apoptosis of hypertrophic chondrocytes and results in disorganization of the growth plate in growing children, a primary cause of the rachitic changes seen at the metaphyses with vitamin D deficiency. The reduced serum levels of calcium and phosphorus also lead to a lower calcium-phosphorus product and the subsequent mineralization defects that are characteristic of rickets. (See ["Overview of rickets in children", section on 'Laboratory findings'.](#))

Skeletal changes

- **Rickets** – Rickets refers to a failure of mineralization of growing bone and cartilage and is the principle manifestation of vitamin D deficiency in infants and young

children. Depending on the severity, the child may be asymptomatic, or have varying degrees of pain and irritability, motor delays, and poor growth. Younger children may manifest with delayed closure of fontanelles, craniotables, frontal bossing, prominence of costochondral junctions, widening of wrists and ankles, and bow legs or knock knees (genu valgum or varum). (See "[Overview of rickets in children](#)".)

Radiologic features of rickets include low bone density; loss of the demarcation between the metaphyses and growth plate and loss of the provisional zone of calcification; widening of the growth plate (from proliferation of uncalcified cartilage and osteoid); and metaphyseal widening, splaying, cupping, and fraying ([image 1](#) and [image 2](#)) [72].

- **Osteomalacia** – Osteomalacia may develop in any age group and is the principle manifestation of vitamin D deficiency in older adolescents and adults. Osteomalacia reflects impaired bone mineralization, and may be asymptomatic or manifest as isolated or generalized muscle and bone pain. Unlike growing children, older adolescents and adults with vitamin D deficiency do not develop rickets or bone deformities, because growth is complete, epiphyseal plates are fused, and there is usually reserve mineral.

Other symptoms — In children with severe vitamin D deficiency, low serum phosphorus levels may cause muscle weakness and discomfort, and children may have difficulty standing or walking.

Patients with advanced vitamin D-deficient rickets occasionally develop seizures or tetany, or may present as apneic spells, stridor, wheezing, hypotonia, and hyperreflexia, particularly in very young children. These symptoms are a consequence of severe hypocalcemia, which is more likely to develop during periods of very rapid growth, such as infancy and adolescence, when increased calcium mobilization from bone from rising levels of PTH and 1,25-dihydroxyvitamin D is unable to keep pace with increased calcium needs. (See "[Clinical manifestations of hypocalcemia](#)".)

APPROACH TO THE DIAGNOSIS

Whom to test

Symptomatic patients — Most patients with vitamin D deficiency are asymptomatic. Symptomatic vitamin D deficiency is most common in infants and toddlers, and consists primarily of skeletal changes (rickets), occasionally accompanied by symptoms caused by hypophosphatemia and/or hypocalcemia (see ['Skeletal changes'](#) above and ['Other symptoms'](#) above). The evaluation of patients with rickets is discussed separately. (See ["Overview of rickets in children", section on 'Evaluation'](#).)

At-risk patients — Screening is recommended in populations at risk for vitamin D deficiency but not for the population at large. We suggest screening the following patient groups, each of which has increased likelihood of rickets or osteopenia [1]:

- Exclusively breastfed or premature infants who are not reliably taking supplements of vitamin D (400 international units [10 micrograms] daily). We suggest screening during a well-infant visit at the time that a risk factor is identified. If the initial result is normal, rescreening is not necessary after infancy, unless there are new risk factors. (See ['Perinatal risk factors'](#) above and ['Targets for vitamin D intake'](#) above.)
- Dark-skinned infants and children who live at higher latitudes, particularly if they also have a history of prematurity. These children should ideally be screened during the winter and spring months, when serum vitamin D concentrations tend to be lowest. (See ['Skin pigmentation and low sun exposure'](#) above.)
- Children with low dietary intake of vitamin D, who are not taking supplements. These children should ideally be screened during the winter and spring months, when serum vitamin D concentrations tend to be lowest. (See ['Decreased nutritional intake'](#) above.)

For children with obesity but no other risk factors, the utility of routine screening is controversial [73]. Some centers routinely screen all children with obesity for vitamin D deficiency because of relatively high rates of vitamin D deficiency in this population. However, some of these cases may be explained by the presence of other risk factors, and the clinical implications of low vitamin D concentrations in this population are unclear.

Screening is also indicated for infants and children with the following special risk factors:

- Infants and young children with nonspecific symptoms such as poor growth, gross motor delays, and unusual irritability. Such symptoms might be caused by rickets or

a condition that predisposes to vitamin D deficiency. (See ['Skeletal changes'](#) above.)

- Children on medications that predispose to vitamin D deficiency, including certain anticonvulsants, antiretroviral drugs, or chronic glucocorticoids. (See ['Medications'](#) above.)
- Children with chronic conditions (see ['Malabsorption and other medical conditions'](#) above):
 - Diseases associated with malabsorption, such as cystic fibrosis, inflammatory bowel disease, or cholestatic liver disease.
 - Other conditions associated with low bone density, such as malnutrition, amenorrhea, or immobilization because vitamin D deficiency may further contribute to reductions in bone density in these conditions.
 - Chronic kidney disease or severe hepatic dysfunction. (See ["Pediatric chronic kidney disease-mineral and bone disorder \(CKD-MBD\)", section on 'Vitamin D deficiency'.](#))
- Adolescents who are pregnant or lactating [\[74\]](#).
- Children with elevated levels of serum alkaline phosphatase for age – eg, >500 international units/L in neonates or >1000 international units/L in children up to nine years of age; alkaline phosphatase levels tend to decrease after puberty [\[75\]](#). (See ['Biochemical changes'](#) above.)

For children with these special risk factors, screening should be performed at the time the risk factor is identified. Subsequent testing depends on the initial result and changes in the medical condition.

How to test — Vitamin D status should be determined by measuring serum 25-hydroxyvitamin D (25OHD). Among the various forms of vitamin D described above, this is the best indicator of vitamin D status and stores. 25OHD is the main circulating form of vitamin D, and has a half-life of two to three weeks. In contrast, 1,25-dihydroxyvitamin D has a much shorter half-life of approximately four hours, circulates in much lower concentrations than 25OHD, and is susceptible to fluctuations induced by parathyroid hormone (PTH) in response to subtle changes in calcium levels.

Most commercial laboratories measure both D₂ and D₃ derivatives of 25OHD, and report the combined result as the 25OHD level. This is important because patients have different proportions of vitamin D₂ and D₃, depending on whether the source is cutaneous synthesis, natural dietary sources, or fortified foods and supplements (see ['Metabolism and forms of vitamin D'](#) above). Reliable assay methods may include a radioimmunoassay, high performance liquid chromatography (HPLC), or liquid chromatography-mass spectroscopy (LC-MS) [76,77]. Variability among assays remains an important problem. (See ["Vitamin D deficiency in adults: Definition, clinical manifestations, and treatment"](#), [section on 'Defining vitamin D sufficiency'](#).)

Diagnosis — Significant controversy has been associated with determining standards of vitamin D sufficiency, insufficiency, and deficiency. Thresholds used to define these states are based upon associations of 25OHD levels with clinical evidence of rickets and elevations in alkaline phosphatase and other bone turnover markers. There is ongoing disagreement about the optimal thresholds because the evidence is inconsistent and because of inconsistency among vitamin D assays.

We suggest the following standards for defining vitamin D status in healthy children and adolescents, based on serum concentrations of 25OHD:

- **Vitamin D sufficiency** – 20 to 100 ng/mL (50 to 250 nmol/L)
- **Vitamin D insufficiency** – 12 to 20 ng/mL (30 to 50 nmol/L)
- **Vitamin D deficiency** – <12 ng/mL (<30 nmol/L)

These standards are consistent with the 2016 Global Consensus recommendations [6], which are similar to 2011 recommendations from the Pediatric Endocrine Society (PES) [1]. They are based upon observations of radiologic changes of rickets and low bone density at 25OHD levels of <16 to 18 ng/mL (40 to 45 nmol/L), and elevations of alkaline phosphatase levels starting around 25OHD levels <20 ng/mL (50 nmol/L) [78-85]. There is little evidence from studies in children to indicate that vitamin D levels above the threshold of 20 ng/mL (50 nmol/L) are necessary to optimize calcium absorption or bone density.

In adults, thresholds for defining vitamin D status are based on associations with PTH levels, and studies of calcium absorption and bone density. Some controversy exists regarding optimal levels [52,55,86,87]. Many experts suggest maintaining 25OHD levels between 20 and 40 ng/mL (50 to 72 nmol/L), while others suggest maintaining 25OHD levels between 30 and 50 ng/mL (75 to 125 nmol/L). The controversy and rationales are

discussed separately. (See "[Vitamin D deficiency in adults: Definition, clinical manifestations, and treatment](#)", section on 'Defining vitamin D sufficiency'.)

Additional evaluation — The possibility of rickets should be considered in growing children with 25OHD levels below 20 ng/mL (50 nmol/L). For these children, the evaluation should include measurements of serum calcium, phosphorus, alkaline phosphatase, and PTH ([table 1](#)). Radiographic evaluation for rickets should be performed if the child is young (eg, <3 years of age) or if there is a high clinical suspicion of rickets, based on risk factors or physical signs. (See "[Overview of rickets in children](#)", section on 'Clinical manifestations'.)

If rickets is present, the results of these laboratory tests can be used to classify the type as calcipenic (hypocalcemic) or hypophosphatemic rickets. The detailed evaluation of a patient with rickets is discussed in a separate topic review. (See "[Overview of rickets in children](#)".)

TREATMENT

Vitamin D deficiency or insufficiency

Vitamin D replacement — Vitamin D replacement therapy is necessary for children presenting with low levels of 25-hydroxyvitamin D (25OHD) <20 ng/mL (50 nmol/L) or rickets. A variety of dosing schemes are used in clinical practice for vitamin D replacement [[1](#)]. Either [vitamin D2](#) (ergocalciferol) or [vitamin D3](#) (cholecalciferol) may be used.

- **Dosing** – In our practice, we use the following doses, which incorporate the Global Consensus recommendations on prevention and management of nutritional rickets [[6](#)]:
 - Infants <12 months old – 2000 international units (50 micrograms) daily for 6 to 12 weeks, followed by maintenance dosing of at least 400 international units (10 micrograms) daily. Commonly available preparations of [vitamin D2](#) are Calciferol or Drisdol oral solution, which provide 8000 international units/mL (200 micrograms/mL).
 - Children ≥12 months old – 2000 international units (50 micrograms) daily for 6 to 12 weeks, followed by maintenance dosing of 600 to 1000 international

units (15 to 25 micrograms) daily. An alternative approach is to treat with 50,000 international units (1250 micrograms) once a week for six weeks [88], followed by maintenance dosing. Although the total dose of vitamin D is higher for the weekly regimen, this approach has been shown to be safe and effective in several trials [89].

- Children with obesity, malabsorptive diseases, or those on medications that impact vitamin D metabolism may require higher replacement doses (two to three times higher than in children without these conditions; thus, as much as 6000 international units [150 micrograms] daily), followed by higher maintenance dosing (see '[Obesity](#)' above and '[Malabsorption and other medical conditions](#)' above and '[Medications](#)' above). Much higher doses may be necessary in conditions such as cystic fibrosis [90]. (See "[Cystic fibrosis: Nutritional issues](#)", [section on 'Vitamin D'](#)".)
- Children with established rickets need somewhat higher treatment doses:
 - Children ≥ 12 months through 12 years old – 3000 to 6000 international units (75 to 150 micrograms) daily
 - Children ≥ 12 years old – 6000 international units (150 micrograms) daily

This is given for 12 weeks, with monitoring for efficacy and the risk of hypercalcemia, followed by maintenance dosing. (See "[Etiology and treatment of calcipenic rickets in children](#)", [section on 'Treatment'](#)".)

Multiple dosing regimens have been shown to be effective. The cumulative amount of vitamin D supplementation appears to be more important than the dosing frequency. As an example, one study in adults found that the same cumulative dose given daily (1500 international units [37 micrograms]), weekly (10,500 international units [262 micrograms]), or monthly (45,000 international units [1125 micrograms]) resulted in similar increments in serum 25OHD concentration [91]. (See "[Vitamin D deficiency in adults: Definition, clinical manifestations, and treatment](#)", [section on 'Dosing'](#)".)

- **Monitoring** – For all patients, serum 25OHD levels should be monitored during or shortly after vitamin D supplementation therapy. The timing and intensity of monitoring depends upon the severity of the deficiency, as discussed below. (See '[Follow-up](#)' below.)

All infants, and those individuals receiving more than 2000 international units (50 micrograms) daily of vitamin D, should be monitored for calcium levels after one to two months to rule out hypercalcemia. This is particularly a possibility in individuals with inactivating mutations in *CYP24A1*. (See '[Prevention in the perinatal period and in infants](#)' below.)

For children who do not achieve therapeutic concentrations of 25OHD following one of the above regimens, the dose of vitamin D should be increased. The specific dose of vitamin D required to raise 25OHD levels into the therapeutic range depends on the severity of the deficiency and individual factors that potentially include vitamin D absorption and degradation of 25OHD. In one study of adults with 25OHD levels <25 ng/mL, 93 percent of those on a dose of 5000 international units (125 micrograms) daily achieved therapeutic levels, compared with 45 percent of those on a dose of 2000 international units (50 micrograms) daily [92]. This study highlights the individual variation in response and emphasizes that even "high" daily vitamin D dosing does not assure achieving a 25OHD concentration of >30 ng/mL.

- **Dosing forms** – Vitamin D may be administered as [vitamin D2](#) (ergocalciferol) or as [vitamin D3](#) (cholecalciferol). The potency of vitamin D3 in relation to vitamin D2 remains somewhat controversial. Typically, the two forms of vitamin D are used interchangeably, particularly with daily dosing [6]. Some studies indicate that vitamin D3 may have a longer half-life than vitamin D2 and may be more potent, causing two- to threefold greater storage of vitamin D [93,94]. Thus, vitamin D3 may be a better option when using a single, large dose [6,95]. Liquid vitamin D preparations containing 8000 international units/ mL of vitamin D2 are available, as are gelatin capsules containing 50,000 international units.

The rare patient with severe symptomatic hypocalcemia due to vitamin D deficiency may benefit from administration of [calcitriol](#) (1,25-dihydroxyvitamin D). In such situations, calcitriol administration at a dose of 20 to 100 ng/kg/day with intravenous [calcium gluconate](#) and high doses of vitamin D may normalize plasma calcium levels more rapidly than standard vitamin D treatments. However, calcitriol plays no role in building up vitamin D stores and should not be used for patients without symptomatic hypocalcemia.

- **Stoss therapy** – Short-term administration of high-dose vitamin D, known as "stoss therapy," is an effective alternative and can be a good solution for patients who do not adhere to oral therapy. Stoss therapy should not be used for young infants (<3 months of age), and careful dosing is important to avoid risks of hypercalcemia [6]. This approach is discussed separately. (See ["Etiology and treatment of calcipenic rickets in children", section on 'Treatment'](#).)

Concomitant calcium supplementation — For patients with elevated levels of parathyroid hormone (PTH) or clinical evidence of rickets, calcium should be supplemented along with vitamin D. This is because vitamin D replacement and a normalization of PTH levels can precipitate hypocalcemia by suppressing bone resorption and from increased bone mineralization, also referred to as the "hungry bone" syndrome. To prevent the hypocalcemia, calcium replacement should be given at doses of 30 to 75 mg/kg/day of elemental calcium, in two or three divided doses. The calcium supplements should be continued for two to four weeks, until vitamin D doses have been reduced to maintenance levels of 600 to 1000 international units daily (see ["Etiology and treatment of calcipenic rickets in children", section on 'Treatment'](#)). The Global Consensus recommendations on prevention and management of nutritional rickets include administration of a daily dose of 500 mg of elemental oral calcium (either dietary or supplemental) with vitamin D treatment, regardless of age or weight [6].

In children with symptomatic hypocalcemia (including seizures or tetany), one or more intravenous boluses of [calcium gluconate](#) may be necessary, at a dose of 10 to 20 mg/kg of elemental calcium (maximum single dose 540 mg) administered slowly intravenously over 5 to 10 minutes (1 to 2 mL/kg of 10 percent calcium gluconate), followed by a slow infusion of calcium in patients with persistent hypocalcemia [96]. (See ["Primary drugs in pediatric resuscitation", section on 'Calcium'](#).)

Follow-up — Patients presenting with only low levels of 25OHD and no other biochemical changes or evidence of rickets do not require intense monitoring. In our practice, we generally check 25OHD levels in such patients after two to three months of vitamin D supplementation therapy, then as needed thereafter, depending on the adequacy of the patient's intake and adherence to maintenance supplements. It is important to monitor 25OHD levels to ensure that vitamin D requirements continue to be met after vitamin D deficiency has been treated, particularly in high-risk populations.

Patients with low vitamin D levels and biochemical changes such as elevated alkaline phosphatase levels or PTH levels, but without rickets, should be monitored more closely to ensure treatment adherence. We generally check serum 25OHD levels and other chemistries after six to eight weeks of high-dose therapy, then again after several months of maintenance therapy, then annually thereafter.

Patients with rickets require close follow-up to document radiographic healing, normalization of serum 25OHD, PTH, calcium and phosphorus levels, and long-term maintenance of vitamin D sufficiency. Recovery is associated with an initial increase in serum phosphate, alkaline phosphatase and 1,25-dihydroxyvitamin D levels, followed by a gradual normalization of these parameters. (See ["Etiology and treatment of calcipenic rickets in children", section on 'Monitoring'.](#))

Borderline vitamin D levels — As discussed above, 25OHD levels above 20 ng/mL have not been associated with adverse clinical effects in children. However, studies in adults have shown impaired calcium absorption and lower bone density at 25OHD levels between 20 and 30 ng/mL (50 to 75 nmol/L), and additional studies are needed to examine these issues more carefully in children. (See ["Diagnosis"](#) above.)

Based on available data, we do not usually give vitamin D replacement therapy to infants or children for low-normal vitamin D levels (eg, 25OHD between 20 and 30 ng/mL [50 to 75 nmol/L]), unless there are other signs of vitamin D deficiency or important risk factors (eg, very low nutritional intake or perinatal risk factors) (see ["Pathogenesis and risk factors"](#) above). However, the diets of such children should be reviewed, and vitamin D supplements should be given as needed to meet intake recommendations. We also suggest monitoring 25OHD levels in these children periodically, and initiating treatment if levels fall below 20 ng/mL (50 nmol/L). (See ["Targets for vitamin D intake"](#) above.)

PREVENTION

Prevention in the perinatal period and in infants

- **Supplementation to the infant** – All exclusively breastfed infants should receive 400 international units (10 micrograms) daily of vitamin D supplements, beginning within a few days after birth [\[1,2,4\]](#). This recommendation is based on the low vitamin D content of breast milk, the inconsistency and unpredictability of

cutaneous vitamin D synthesis from sun exposure, and the disproportionately high frequency of rickets among exclusively breastfed infants. Supplementation should be continued until the infant is weaned and drinks at least 33 ounces (1 liter) of vitamin D-fortified formula (or vitamin D-fortified cow's milk or fortified plant-based milk, if infant is older than 12 months). Other sources of vitamin D include sunlight and certain solid foods (oily fish, eggs, and fortified foods). However, the intake from these sources tends to be low and inconsistent, so it is best to rely on either supplements or vitamin D-fortified formula/cow's milk to supply vitamin D requirements (see ['Exclusive breastfeeding'](#) above). For premature infants, most authorities in the United States recommend supplementation with 400 international units (10 micrograms) daily (similar to term infants). Higher doses are recommended in European and Canadian guidelines. (See ["Management of neonatal bone health", section on 'Vitamin D'](#).)

Many formula-fed infants should also receive vitamin D supplements. Fortification practices in the United States ensure that infant formulas contain 40 to 100 international units (1 to 2.5 micrograms) of vitamin D per 100 kcal of formula, providing at least 400 international units (10 micrograms) per liter. Thus, formula-fed infants who consume at least 1 liter (33 oz) of formula daily meet the American Academy of Pediatrics (AAP) standards for vitamin D intake. However, most infants who are only partially formula-fed and many infants who are fully formula-fed will consume less than this amount of formula and should therefore receive supplemental vitamin D.

Of concern, fewer than one-third of infants in the United States are receiving sufficient vitamin D to meet the AAP recommendations [\[97\]](#). In a national survey, only 20 percent of breastfeeding infants and 31 percent of nonbreastfeeding infants met the AAP target for vitamin D intake. This is partly because pediatric health care providers in the United States are not routinely advising vitamin D supplements for predominantly breastfed infants. In one study, only 36 percent of responding clinicians indicated that they routinely recommended vitamin D supplementation in predominantly breastfed infants [\[98\]](#). In addition, an even smaller percentage of parents are actually giving vitamin D supplements to their infants. In the study cited above, 67 percent of parents indicated that they believed breast milk has all necessary nutrients, and only 3 percent gave supplements to their children [\[98\]](#).

Awareness of and adherence to national recommendations for vitamin D supplementation also appears to be a problem in the United Kingdom [99-102]. Adherence is better in some other countries. In a Canadian study, 74 percent of mothers who exclusively breastfed their infants indicated compliance with Canadian recommendations for vitamin D intake (also 400 international units [10 micrograms] daily) [103]. Other reports describe that supplements are given as recommended to 59 percent of breastfed infants in Norway and 64 percent of those in Sweden [104,105].

Rarely, standard vitamin D supplementation may trigger **idiopathic infantile hypercalcemia** (IIH), which is characterized by hypercalcemia, failure to thrive, vomiting, dehydration, and nephrocalcinosis. The disorder has been attributed to mutations in *CYP24A1*, which encodes the primary enzyme responsible for degradation of both 25OHD and 1,25-dihydroxyvitamin D [106]. The effect is dose-related, and the disorder is uncommon among infants given standard supplement doses of vitamin D. The risk and speed of developing symptomatic IIH appears to be greater in infants given bolus dosing of vitamin D (eg, 600,000 international units [15,000 micrograms] every three months, as has been done in some countries).

The lack of information about the frequency of *CYP24A1* mutations precludes a universal screening recommendation for IIH in infants given standard doses of vitamin D supplements. However, infants should be evaluated for the possibility of IIH if they develop suspicious symptoms or have a family history of hypercalcemia. The first step in the evaluation is to measure serum 25-hydroxyvitamin D (25OHD) and calcium levels. If these levels are elevated (eg, 25OHD >50 ng/mL [72 nmol/L], and calcium >the upper limit of normal for age), then vitamin D supplements should be stopped and the infant should be further evaluated for evidence of IIH, which includes suppressed parathyroid hormone (PTH), hypercalciuria, and nephrocalcinosis.

- **Supplementation to the lactating mother** – The administration of moderately high doses of vitamin D (4000 to 6400 international units [100 to 160 micrograms] daily) to the lactating mother is another strategy to raise vitamin D levels in exclusively breastfed infants without giving supplements to the infant [5,107,108]. This intervention increases the vitamin D content of breast milk to allow sufficient vitamin D intake by the infant. In a randomized trial, equivalent infant vitamin D

status was achieved by providing supplements of 6400 international units/day (160 micrograms/day) to the lactating mother, compared with supplements of 400 international units/day (10 micrograms/day) to the infant [108]. There was no evidence of vitamin D toxicity (eg, hypercalciuria) in the mothers who were taking the 6400 international units/day (160 micrograms/day) supplement.

Lower vitamin D doses (eg, maternal intake of 1500 to 2400 international units [37.5 to 60 micrograms] daily during pregnancy and lactation) may not result in sufficient vitamin D in breast milk to meet the infant's needs, and supplementation may still be necessary for the infant, although they are generally sufficient to maintain serum levels of vitamin D of >30 ng/mL in the mother and will improve the infant's vitamin D status at birth. Similarly, supplements of 2400 international units/day (60 micrograms/day) given to lactating mothers were not sufficient to prevent vitamin D insufficiency in unsupplemented infants [108].

- **Vitamin D supplementation of pregnant women** – To optimize an infant's vitamin D status and bone health at birth, it is important to ensure that the pregnant mother has sufficient vitamin D intake throughout pregnancy. This is because maternal vitamin D crosses the placental barrier and builds up fetal stores of vitamin D, particularly during the third trimester. This is of greater concern in dark-skinned women, those living in higher latitudes, and those whose cultural and religious practices include complete skin cover.

In pregnant and lactating women, the recommended dietary allowance for vitamin D is 600 international units (15 micrograms) daily, which is the same as for women who are not pregnant [2]. However, some studies suggest that this intake may not be adequate. One study of pregnant women in Finland found that 71 percent were vitamin D deficient (25OHD levels <20 ng/mL) despite an average vitamin D intake of almost 600 international units daily; lower maternal 25OHD levels were also associated with some indicators of reduced fetal bone health [109]. Other studies suggest that doses of vitamin D in excess of 1000 international units per day are necessary to achieve 25OHD concentrations of >20 ng/mL (50 nmol/L) in pregnant women, particularly those with dark skin pigmentation [110-116]. (See "[Vitamin D deficiency in adults: Definition, clinical manifestations, and treatment](#)", [section on 'Pregnancy'](#).)

Effects of maternal vitamin D supplementation during pregnancy on bone health in the offspring are discussed separately. (See ["Vitamin D and extraskkeletal health", section on 'Pregnancy outcomes'](#).)

Prevention in older children and adolescents

- **Vitamin D fortification of milk and other foods** – In the United States, milk and some brands of orange juice are fortified with 100 international units (2.5 micrograms) of vitamin D per cup. Consumption of at least one liter of fortified formula or beverages daily is usually sufficient to meet at least two-thirds of the more current guidelines for daily vitamin D intake (600 international units daily [15 micrograms] for children one year and older). However, many children do not consume this quantity of fortified beverages and may need supplementation to meet guidelines for vitamin D intake. This is particularly true if juice intake is limited because of its high content of sugar and calories, which have been implicated in the development of childhood obesity.

Milk is not routinely fortified with vitamin D in many countries outside of the United States. Fortification practices and vitamin D intakes vary widely among European countries [117], and nearly 45 percent of children and adolescents across Europe have vitamin D insufficiency or deficiency (serum 25OHD <20 ng/mL [50 nmol/L]) [118]. There is ongoing controversy about optimal strategies to address this problem. On the one hand, a supplementation strategy does not reach the entire population, because of nonadherence. On the other hand, a milk-fortification strategy does not ensure adequate intake, because milk intake tends to vary widely within a population. Several studies have suggested that milk fortification has only modest effects on the prevalence of vitamin D insufficiency and deficiency [119,120].

- **Exposure to sunlight** – Sun exposure allows for cutaneous vitamin D synthesis. In individuals with light skin pigmentation, during most seasons 10 to 15 minutes of sun exposure near midday is sufficient for adequate vitamin D synthesis [1]. However, darker skin pigmentation, liberal use of sunblock, winter season, or northern latitudes can markedly reduce skin synthesis of vitamin D and increase the need for dietary sources. (See ['Skin pigmentation and low sun exposure'](#) above.)

Although sun exposure allows cutaneous vitamin D synthesis, we do not encourage sun exposure for prevention of vitamin D deficiency, given the associated risk of skin cancer. These concerns have led to recommendations that direct sunlight exposure should be avoided for infants younger than six months old, and that sun exposure should be limited in older children, through the use of protective clothing and sunscreen [121,122]. Studies are necessary to assess the impact of these recommendations in dark-skinned children, and it is possible that relaxation of these measures in dark-skinned children will allow for sufficient cutaneous vitamin D synthesis in the summer months, particularly in lower latitudes. (See ["Primary prevention of melanoma"](#).)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Vitamin D deficiency"](#) and ["Society guideline links: Pediatric bone health"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topic (see ["Patient education: Vitamin D for babies and children \(The Basics\)"](#))

SUMMARY AND RECOMMENDATIONS

- The recommended intake for vitamin D is 400 international units daily (10 micrograms) for infants, beginning soon after birth, and 600 international units (15 micrograms) for children and adolescents 1 to 18 years of age. High-risk groups may have a higher requirement of vitamin D to maintain serum 25-hydroxyvitamin D (25OHD) levels in the sufficient range. (See ['Targets for vitamin D intake'](#) above.)
- Groups at risk for vitamin D deficiency include:
 - Exclusively breastfed infants (unless they are reliably taking supplements of 400 international units daily)
 - Premature infants
 - Dark-skinned children on diets with low vitamin D content (eg, low dairy intake and some vegetarian and unusual diets, including a vegan diet)
 - Children living at higher latitudes
 - Children with conditions of malabsorption or those who are taking certain medications(See ['Pathogenesis and risk factors'](#) above.)
- For infants and children with the above risk factors, we suggest laboratory screening for vitamin D deficiency ([Grade 2C](#)). Screening is accomplished by measuring serum concentrations of 25OHD. The timing of such screening depends on the underlying risk factor. Infants who are reliably taking the recommended supplements do not require routine laboratory screening for vitamin D deficiency. (See ['At-risk patients'](#) above.)
- Standards for defining vitamin D status in healthy children are not well established. The most widely accepted definitions are (see ['Diagnosis'](#) above):
 - **Vitamin D sufficiency** – 25OHD ≥ 20 ng/mL (50 nmol/L)
 - **Vitamin D insufficiency** – 25OHD 12 to 20 ng/mL (30 to 50 nmol/L)
 - **Vitamin D deficiency** – 25OHD < 12 ng/mL (< 30 nmol/L)
- For infants and children with 25OHD concentrations below 20 ng/mL (50 nmol/L), we recommend vitamin D repletion ([Grade 2C](#)). Children with 25OHD concentrations below this threshold are at increased risk for radiologic changes of

rickets and low bone density, and vitamin D supplementation is generally safe and effective. In our practice, we use a 6- to 12-week course of vitamin D replacement at doses ranging from 2000 to 6000 international units (50 to 150 micrograms) daily, depending on the degree of deficiency and the age of the individual, followed by maintenance dosing of 400 to 1000 international units (10 to 25 micrograms) daily. Some children may require higher doses. Either [vitamin D2](#) (ergocalciferol) or [vitamin D3](#) (cholecalciferol) may be used. A variety of other dosing schemes are also effective for children older than one year, including use of 50,000 international units (1250 micrograms) given once weekly for six or more weeks. (See '[Vitamin D replacement](#)' above.)

- After treatment for vitamin D deficiency, follow-up laboratory testing is important to verify response and adherence to treatment and to verify that normal vitamin D levels are sustained on maintenance dosing. (See '[Follow-up](#)' above.)
- For exclusively breastfed infants, we recommend vitamin D supplementation providing 400 international units (10 micrograms) daily (**[Grade 1B](#)**). Infants who are partially formula-fed usually also require supplementation, unless their formula intake is >1000 mL (33 oz) daily. Use of supplements in exclusively breastfed neonates and infants may be avoided if maternal intake of vitamin D is 4000 to 6000 international units (100 to 150 micrograms) daily. Supplementation should be continued until the infant is weaned and drinks at least 33 ounces (1 liter) of vitamin D-fortified formula (or fortified cow's milk, if the infant is older than 12 months). (See '[Prevention in the perinatal period and in infants](#)' above.)

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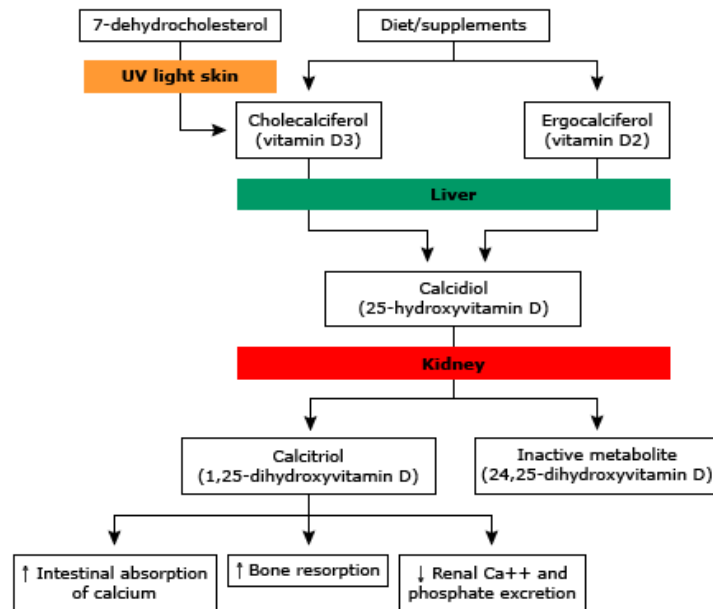
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GRAPHICS

Pathways of vitamin D synthesis



Metabolic activation of vitamin D to calcitriol and its effects on calcium and phosphate homeostasis. The result is an increase in the serum calcium and phosphate concentrations.

UV: ultraviolet; Ca: calcium.

Graphic 65360 Version 6.0

Biochemical manifestations of different stages of vitamin D deficiency, as compared with deficiencies of calcium or phosphorus

	Plasma Ca ⁺⁺	Plasma PO ₄	ALP	PTH	25(OH)- D	1,25(OH) ₂ - D	X-ray changes
Vitamin D deficiency							
Early	N or ↓	N or ↓	↑	↑	↓	N	Reduced bone mineralization
Moderate	N or ↓	↓	↑↑	↑↑	↓↓	↑	Rachitic changes +
Severe	↓↓	↓↓	↑↑↑	↑↑↑	↓↓↓	↑ or N or ↓	Rachitic changes ++
Calcium deficiency	N or ↓	↓	↑↑	↑	N	↑	
Phosphorus deficiency	N or ↑	↓↓	↑↑	N or ↓	N	↑	

N: normal; ALP: alkaline phosphatase; PTH: parathyroid hormone; 25(OH)D: 25-hydroxyvitamin D; 1,25(OH)₂D: 1,25-dihydroxyvitamin D.

Data from: Levine MA, Zapalowski C, Kappy MS. Disorders of calcium, phosphate, parathyroid hormone, and Vitamin D. In: Kappy MS, Allen DB, and Geffner ME (Eds). *Principles and Practice of Pediatric Endocrinology*. Charles C. Thomas Co, Springfield, 2005.

Graphic 56687 Version 5.0

Vitamin D deficiency rickets in a child

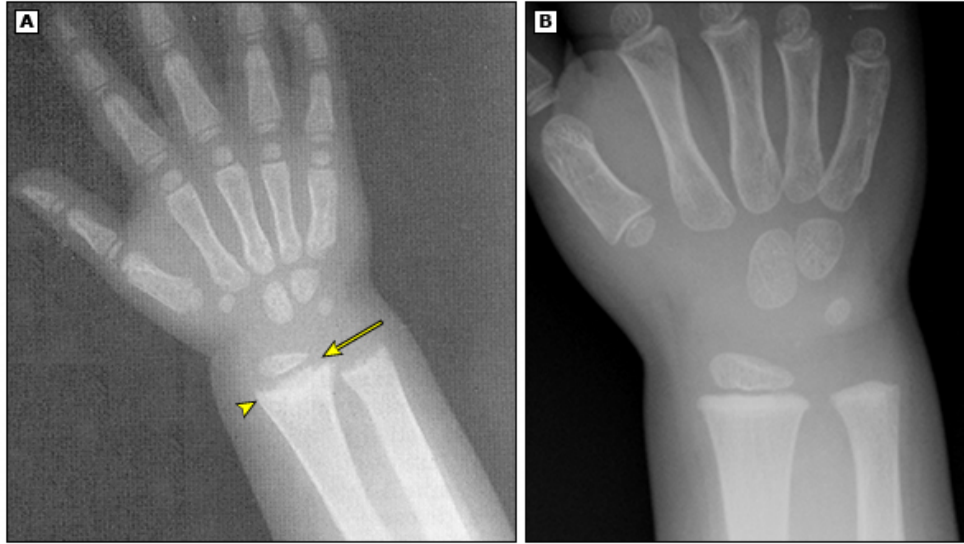


Characteristic findings of rickets in children often include radiographic evidence of decreased mineralization around the epiphyses and bowing of the lower extremities.

<http://www.asbmr.org>.

Graphic 53635 Version 2.0

Anteroposterior radiograph of the wrist and hand in a child with rickets



(A) Rickets. Anteroposterior radiograph of the wrist and hand in a 3-year-old child with nutritional rickets. The child had been put on a strict diet without dairy products. Note the widening, cupping, and fraying of the distal radius (arrowhead) and ulna metaphyses with an associated increase in the thickness of the growth plate (arrow). These changes are the consequence of disordered endochondral growth.

(B) Normal. Radiograph of the hand of a healthy 3-year-old child, without rickets.

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Panel B courtesy of: Lachlan Smith, MD.

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